

HS Code Crosswalk for Pharmaceutical Goods

Objective

This document describes the three-stage concordance built to solve that, producing a single stable identifier (family_id) per commodity that survives all code changes from 1988 through 2026.

Key concepts and column definitions

Every row in lookup.csv corresponds to exactly one source-CSV Commodity string. The 11 columns (user-simplified schema, 2026-05-14):

Column	Meaning
uid	Stable per-row identifier in the form IMR_0001 (imports) or EXR_0001 (exports). Assigned deterministically by sorting orig_commodity alphabetically and numbering sequentially. Use this instead of the long descriptive string when referencing a specific row in conversation. Stable across pipeline re-runs as long as the source CSV's set of distinct Commodity strings doesn't change.
orig_commodity	The exact Commodity field from the source CSV (preserves the Canadian shorthand intact so existing joins on the source still work). The join key when matching back to monthly trade data.
orig_hs_code	The HS code parsed from this row's commodity string.
orig_description	The description parsed from this row's commodity string, with Canadian shorthand expanded to full words for readability — e.g., nes → "not elsewhere specified", therap → "therapeutic", prophltc → "prophylactic", o/t No 30.02/05/06 → "other than products of headings 30.02, 30.05, 30.06". (Termination marker stripped.) The exact source-CSV form is preserved in orig_commodity for joins.
orig_first_period	Earliest activity of THIS specific Commodity string (per-variant). For a silently-retired variant in a still-active family, this shows the variant's own start.
orig_last_period	Latest activity of THIS specific Commodity string. For a silently-retired variant, this is when its data actually stopped (e.g., 2001-12 for 3001.90.00.90 - Other H/A subs ... even though the family containing the active Heparin variant runs to 2025-12).
family_id	Permanent commodity-family identifier (e.g., IM_0008). Stable across revisions of this dataset — the durable primary key for any database or web app.
latest_commodity	The family's canonical source-CSV string (the latest active variant's Commodity row). Same format as orig_commodity — read them in parallel: orig_commodity is where this row started, latest_commodity is where the family is now.
latest_hs_code	The family's currently-active HS code (or last-active if the family is dead).
latest_description	Display description matching latest_hs_code, also shorthand-expanded.
latest_status	active if any family member is still trading; ended_YYYY-MM if all members are terminated. Reflects the silent-retirement inference — variants without a source (Terminated ...) tag are still marked ended_* when their data has stopped and another variant under the same code has taken over.

Companion files

Two complementary views ship alongside lookup.csv:

- `lineage_timeline.csv` (10 columns) — one row per **step** in a commodity's forward chain. Shows the per-step `hs_code`, the step's description (shorthand-expanded), `orig_first_period` / `orig_last_period`, `step_status` (active / ended_YYYY-MM), and `next_hs_code`. Useful for auditing how each commodity migrated across HS revisions step by step.
- `commodity_journeys.csv` (10 columns) — one row per `orig_commodity` (same UID-set as `lookup.csv`) with the entire forward chain rendered into a single `journey` cell as HS1 -> HS2 -> HS3. Includes `orig_status` (this variant's own state), `n_steps_in_chain`, `latest_commodity` (the family's canonical), and `final_status` (the family's current state — same value as `lookup.latest_status`).

Canadian Customs description shorthand

Canadian HS sub-code descriptions use heavy abbreviation. Reading them quickly is the single biggest skill for sanity-checking a routing.

Shorthand	Decoded meaning
o/t	Other than (most often misread)
o/t No 30.02/05/06	Other than products of headings 30.02, 30.05, 30.06 — appears in residual sub-headings to exclude codes that have their own home
nes	Not elsewhere specified — the universal Canadian residual marker
frs	For retail sale — implies in dosage form, packaged for retail (3004.x range)
nfrs	Not for retail sale — implies in bulk (3003.x range)
cont	Containing
Medi, Med	Medicaments
mix	Mixed (mixture of substances)
therap/prophltc	Therapeutic / prophylactic
deriv	Derivatives
t/o	Thereof (their derivatives etc.)
hum	Human
vet	Veterinary
horm	Hormones
prep	Preparations
sim	Similar
mfd	Manufactured
o/w	Otherwise

bld	Blood
fract	Fractions
parentl / parenteral	Administered by injection (not by mouth)
anml, aml, vet	Animal / veterinary

A worked example of decoding:

Medi,cont alkaloids/deriv t/o,nes,therap/prophlhc,doses/frs,o/t No 30.02/05/06

>

Medicaments, containing alkaloids or their derivatives thereof, not elsewhere specified, therapeutic / prophylactic uses, in dosage / for retail sale, other than products of headings 30.02, 30.05, 30.06.

In other words: the post-HS2017 alkaloid catch-all for in-dosage pharmaceutical products. HS6 3004.49.

The Three-Stage Approach

Stage 1 — Linking from CSV data only

For each terminated code T, the script looks for a successor code N within the inventory such that:

- N's first activity period falls within [T's termination month, T's termination month + 12 months].
- The text similarity between T's and N's descriptions is ≥ 0.85 (using Python's `difflib.SequenceMatcher` on lowercase, punctuation-stripped strings).

If such an N exists, T and N are merged into the same family (`family_id`).

Before Stage 1's windowing, every `raw_commodity`'s `terminated_yyyymm` is recomputed. A variant is silently retired at its own `last_period` if some other variant under the same HS code has `first_period > last_period`. This catches two situations:

1. **No source tag** — Statistics Canada relabelled the description at an HS revision boundary without applying a (Terminated ...) marker. Example: 3001.90.00.90 - Other human or animal subs prepared for therapeutic or prophylactic uses, nes runs from 1998-01 to 2001-12; the same code is then re-described in 2002 as Heparin & its salts; human or animal substances for therap or prophlhc uses, nes. Before the fix, the old wording stayed marked active forever. After the fix, it reads ended_2001-12.
2. **Wrong source tag** — Statistics Canada applies the (Terminated YYYY-MM) tag at the *code* level, so every description-variant of a retired code carries the same date even when it doesn't reflect when the specific variant actually stopped. Example: 3004.10.00.13 - Penicillin V for human use, in dosage carries (Terminated 2011-12) but its data dies 1991-10 when Amoxycillin moves into the same suffix. Without this correction, Stage 1 windows from the wrong year and misses the natural same-drug cross-suffix link.

Stage 1 found 7 successor pairs in exports and 189 in imports (the imports figure rose from 147 pre-fix because the silent-retirement detection unlocks correctly-windowed Stage-1 candidates that previously fell through to Stages 2 or 3).

Examples:

- 3001.90.10.00 Heparin (1988-1997) → 3001.90.00.10 Heparin (1998-present) — identical description.
- 3002.20.00 Vaccines for human medicine → 3002.41.00 Vaccines, for human medicine (HS2022).

Stage 2 — Human review of moderate-similarity candidates

There are instances where human interference was needed because the script misses cases where the description was reworded but the commodity is clearly the same.

Example: Blood plasma's 2016-12 → 2017-01 transition reworded "Blood plasma, for therapeutic or prophylactic use in humans" to "Human blood plasma, for therapeutic/prophylactic/other uses".

Stage 2 in this dataset: 5 exports rows approved (out of 5 review candidates); 50 imports approved + 76 imports "difficult" out of 126 review candidates

Stage 3 — Domain-knowledge routing for unresolved cases

Stage 3 takes the rows that Stage 1 couldn't auto-link and Stage 2 didn't approve, and routes them using:

- World Trade Organization HS6 transition documents — a table of known international HS6 transitions for chapter 30.
- Same-HS6 active catch-all detection — within a given HS6, the residual sub-code is identified by Canadian convention: descriptions containing nes, suffixes ending in 9 (.X9), or .90 / .99. The script also handles HS6s where only one active code per human/vet context exists (e.g., 3004.10 has just .10 for human and .90 for vet).
- Human/vet context routing — if the description says "for human use", the router prefers active codes whose description contains "human" or ,hum,. Symmetric for vet/veterinary.

Stage 3 in this dataset: 5 exports routings and 107 imports routings.

Example 1 — Blood plasma (1988–2026, three generations of HS10 codes)

HS10	Period	Termination	Stage that found successor
3002.10.10.21	1988-01 → 1997-12	1997-12	Stage 1 — auto-linked to 3002.10.00.21 (identical description)
3002.10.00.21	1998-01 → 2016-12	2016-12	Stage 2 — analyst-approved link to 3002.12.00.12 (description similarity)
3002.12.00.12	2017-01 → present	(active)	active_never_terminated

All three rows share `family_id = IM_0008`. A monthly series joined on `family_id` is continuous from 1988-01 to 2025-12 with no zero-cliffs at the 1998 or 2017 boundaries.

Example 2 — Vaccines for veterinary medicine (Canadian merge + WCO rename)

HS8	Period	Termination	Stage
3002.30.10	2000-04 → 2016-06	2016-12 (foot-and-mouth-disease vaccines)	Stage 2 — analyst-approved merge into 3002.30.00
3002.30.90	2000-01 → 2016-12	2016-12 (vaccines vet, nes)	Stage 2 — analyst-approved merge into 3002.30.00
3002.30.00	2017-01 → 2021-12	2021-12 (Canadian consolidation in 2017)	Stage 1 — auto-linked to 3002.42.00
3002.42.00	2022-01 → present	(active)	active_never_terminated

Three distinct events stitched together: a 2017 Canadian consolidation (.10 and .90 Canadian tariff items collapsed into .00), and the 2022 WCO HS6 rename (3002.30 → 3002.42). All four rows are in one family, one continuous time series.

Code-change patterns across the dataset

We observed the following patterns across the dataset:

Pattern 1 — Continuously active (no change)

Some HS codes were introduced once and have been in continuous use without renumbering or termination.

- **Mechanism:** no transitions needed. `successor_stage = active_never_terminated` for every row.
- **Counts:** 22 imports families, 21 exports families.
- **Example:** 3001.20.00.00 Extracts of glands or other organs or of their secretions, for therapeutic uses — active continuously from 1988 to 2026.

Pattern 2 — Description revision under the same code

Same HS code, same scope, but the description was reworded over time (typo fixes, abbreviation changes, wording updates).

- **Mechanism:** same HS code automatically merged by Rule 0 (the default, non-overridden case).
- **Counts:** 3 imports families, 13 exports families.
- **Example:** 3001.90.00 Heparin & its salts; human or animal substances (typo present 2000-2006) was corrected to Heparin & its salts; human or animal substances in 2007. Same HS code throughout; merged into one family. The old wording's `audit_silent_retirement`: Y annotation marks the moment Stats Canada switched the spelling.

Pattern 2b — Code repurposed for a different commodity (Rule-0 split)

New in 2026-05-14. Same HS code, but the suffix was reused over time for genuinely different commodities. Rule 0's blanket merge would contaminate the time series; the surgical `SPLIT_OVERRIDES` table forces these into separate families.

- **Mechanism:** Rule-0 split + cluster-aware fusion of near-identical variants. Each cluster becomes its own Stage-1 root; Stage 1's auto-linker is forbidden from re-merging cross-cluster pairs and from cross-HS6 mismerges of the split orphans.
- **Counts:** 5 imports HS codes (12 raw_commodity strings); 0 exports.
- **Audit flag:** `audit_rule0_split` = Y for every variant of every override-flagged code.

The five override-flagged codes (imports):

HS code	Pre-fix description / period	Post-fix description / period
3002.90.00.20	Butter-making ferment cultures (1998-2016)	Toxoids, antitoxins, viruses, bacteriophage, bacterial lysates (2022+)
3004.10.00.12	Penicillin G procaine (1988-1991)	Penicillin V (1992-2011)
3004.10.00.13	Penicillin V (1988-1991)	Amoxycillin (1992-2011)
3004.20.00.64	Rifamycins (1988-1991)	Lincomycin (1991-12)
3004.50.99.30	Vitamins nes for vet use (1988-1989)	Multivitamins not mixed, for human use (1990-1997)

The butter cultures cluster has no chapter-30 successor and is routed via a Stage-3 direct override to `no_successor_found` with reason "industrial dairy/food fermentation cultures, proper HS classification in chapter 4 or 21".

Pattern 3 — Simple rename, description preserved

Statistics Canada renumbered the Canadian sub-code at a customs-restructure boundary, but the description remained identical or nearly identical.

- **Mechanism:** Stage 1 auto-link (similarity ≥ 0.85 , within 12 months).
- **Counts:** 33 imports families, 2 exports families.
- **Example:** 3001.90.10.00 Heparin and its salts (1988-1997) → 3001.90.00.10 Heparin and its salts (1998-2026). Identical description; only the Canadian tariff-item digits changed (.10.00 → .00.10).

Pattern 4 — Renumber with description reworded enough to need review

The HS code changed AND the description was reworded enough to require analyst judgment at Stage 2.

- **Mechanism:** Stage 2 user approval (review candidate confirmed).
- **Counts:** 1 imports family, 1 exports family resolved by Stage 2 alone.
- **Example:** 3002.41.00.10 Vaccines, for against influenza virus, human medicine (HS2022 vintage) merges with 3002.20.00.10 Vaccines, for human use against influenza virus from earlier years.

Pattern 5 — Consolidation into a post-2012 catch-all

Multiple specific Canadian sub-codes from before 2012 funnel into a single post-2012 catch-all code. The dominant pattern produced by the 2012 Canadian restructure.

- **Mechanism:** Stage 3 domain routing using catch-all detection rules.
- **Counts:** 2 imports families, 1 exports family classified primarily as this pattern; many Pattern 6 cases also exhibit this internally.
- **Example:** 16 specific antibiotic Canadian sub-codes (Tetracyclines .10, Macrolide nes .39, Gentamicins .41, Bacitracin .51, Polymyxins .52, Polypeptide nes .59, Chloramphenicol .61, Mitomycin C .62, Lincomycin .64/.65, Antibiotics nes .69, etc.) all consolidated into 3004.20.00.19 Medi cont antibiotics, hum, nes post-2012.

Pattern 6 — Multi-step chain across multiple stages

The family went through several code changes across the 1988–2026 window, with different stages of the process resolving each transition. The most complex and most common pattern in imports.

- **Mechanism:** combinations of Stage 1 + Stage 2 + Stage 3 transitions.
- **Counts:** 44 imports families, 2 exports families.
- **Example:** Blood plasma family IM_0008 — see Example 1.

Pattern 7 — Cross-chapter exit

The commodity left chapter 30 entirely via an HS revision. Most commonly the HS2022 move of diagnostic reagents to chapter 38.

- **Mechanism:** marked as no_successor_found in lookup_3; no_successor_reason documents the chapter-38 destination.
- **Counts:** 3 imports families, 2 exports families.

Pattern 8 — Out of scope (never really chapter 30)

The commodity was historically classified in chapter 30 but properly belongs in another chapter.

- **Mechanism:** marked as no_successor_found; no_successor_reason indicates the proper destination.
- **Counts:** 2 imports families (Bacteria culture inoculants → chapter 31; Butter-making ferment cultures → chapter 4/21), 0 exports families.

Results summary

Metric	Exports	Imports
Total commodity rows	78	475
Distinct families after Stage 3	48	118
successor_found rows	76	470
no_successor_found rows	2	5
Found at active_never_terminated	46	115
Found at Stage 1 (auto-link)	20	211
Found at Stage 2 (human approval)	5	44
Found at Stage 3 (domain routing)	5	100
No successor (none)	2	5